

THE ZERO PARADOX

ZP-J Wheel Addendum

The Wheel of Fractions is a Wheel

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A wheel (Carlström 2001) is an algebraic structure that extends a commutative ring by making division a total operation: every element, including 0, has a reciprocal $/x$, so $/0$ becomes a defined first-class element rather than an error. The two elements this produces — $\infty = /0$ (the reciprocal of zero) and $\perp = 0 \cdot /0$ (an absorbing "undefined" element) — are what distinguish a wheel from a field. ZP-J Self-Reference left open which of the two standard ways of totalising division the Zero Paradox porthole gives rise to: a wheel, which adjoins ∞ and $\perp$ as two distinct new elements, or a meadow (Bergstra & Tucker), which adjoins nothing at all and instead declares $0^{-1} = 0$, staying inside the original number system. This addendum settles that question.

The main result is `WheelFrac.instWheel` (`WheelFrac.lean`): for any commutative ring A and any multiplicative submonoid S , the wheel of fractions $\odot_S A = (A \times A) / \equiv_S$ satisfies every axiom of Carlström's Definition 1.1. The companion result `WheelFrac.inf_ne_bot` shows that, whenever $0 \notin S$, the two special elements stay distinct ($\infty \neq \perp$) — so the construction is a non-trivial wheel. The construction is Carlström's; the contribution here is a faithful, machine-verified encoding of it that is also free of the axiom of choice (footprint `[propext, Quot.sound]`), situated as the algebraic form of the ZP porthole.

Section I: The Wheel Axioms (Carlström Definition 1.1)

A wheel is a set W with two binary operations $+$ and $\cdot$, a unary involution $/$ (reciprocal), and two constants 0 and 1 , subject to eight axioms. Carlström's Definition 1.1 packages the additive and multiplicative parts as commutative monoids; the ZP Wheel typeclass unbundles those two monoid axioms into their separate equational laws, giving 14 fields that are equivalent to Carlström's eight. The unbundling is bookkeeping only — no axiom is added, removed, or weakened.

Typeclass: `Wheel` (`Wheel.lean`)

`class Wheel (W : Type*) where`

`wadd, wmul : W → W → W -- + and ·`

`winv : W → W -- the involution /`

`wzero, wone : W -- 0 and 1`

`-- W1–W3: (W, +, 0) commutative monoid [Carlström (1)]`

Typeclass: Wheel (Wheel.lean)

-- W4–W6: $(W, \cdot, 1)$ commutative monoid [Carlström (2), monoid part]

-- W7: $/(/x) = x$ [Carlström (2), involution]

-- W8: $/(x \cdot y) = /x \cdot /y$ [Carlström (2), involution]

-- W9: weakened distributivity [Carlström (3)]

-- W10: $(x \cdot /y + z) + 0 \cdot y = (x + y \cdot z) \cdot /y$ [Carlström (4)]

-- W11: $0 \cdot 0 = 0$ [Carlström (5)]

-- W12: $(x + 0 \cdot y) \cdot z = x \cdot z + 0 \cdot y$ [Carlström (6)]

-- W13: $/(x + 0 \cdot y) = /x + 0 \cdot y$ [Carlström (7)]

-- W14: $x + 0 \cdot /0 = 0 \cdot /0$ [Carlström (8)]

The last group of axioms is what makes division total. W7 and W8 make $/$ an involution that distributes over multiplication; W9–W14 govern how the two derived elements $\infty = /0$ and $\perp = 0 \cdot /0$ interact with $+$ and $\cdot$. Keeping those two apart is not a stylistic preference over some rival algebra: by Carlström’s Proposition 4.4 (p. 25), if any two of 0 , 1 , $/0$ and $0/0$ are equal in a wheel, the wheel is trivial — it has exactly one element. So $\infty \neq \perp$ is the condition for the structure to have more than one element at all. The same statement is machine-checked over the Wheel typeclass in ZeroParadox/Algebra/Wheel.lean §V: identifying ∞ with $\perp$ proves $\forall x y, x = y$.

Section II: The Wheel of Fractions $\odot_S A$

The wheel of fractions generalises the field-of-fractions construction so that it survives division by zero. Start from a commutative ring A and a multiplicative submonoid $S \subseteq A$ (a set containing 1 and closed under multiplication). Form pairs $(x, y) \in A \times A$ — read as the formal fraction x/y — and quotient by the relation $\equiv_S$ below.

Construction (Carlström 2001, pp. 4–5)

$$\odot_S A = (A \times A) / \equiv_S$$

$$(x, y) \equiv_S (x', y') \Leftrightarrow \exists s, s' \in S, s \cdot x = s' \cdot x' \wedge s \cdot y = s' \cdot y'$$

$$0 = [0, 1] \quad 1 = [1, 1]$$

$$[x, y] + [x', y'] = [x \cdot y' + x' \cdot y, y \cdot y']$$

$$[x, y] \cdot [x', y'] = [x \cdot x', y \cdot y']$$

$$/[x, y] = [y, x] \text{ (the involution is pair-swap)}$$

Construction (Carlström 2001, pp. 4-5)

Then $1/0 = [1,0] = \infty$ and $0/0 = [0,0] = \perp$.

The choice of relation matters. The naive cross-multiplication $x \cdot y' = x' \cdot y$ — the one that defines equality of ordinary fractions — is not an equivalence relation on a general commutative ring: transitivity fails without a cancellation law. The submonoid-quotient relation $\equiv_S$ repairs this by witnessing each identification with elements of S , and it is provably reflexive, symmetric, and transitive (`WheelFrac.srel`). Each of the five operations is then well-defined on the quotient — the proofs that they respect $\equiv_S$ are the bulk of the formalisation.

Section III: $\odot_S A$ is a Wheel

With the operations shown well-defined, every one of the 14 typeclass fields is discharged. The additive and multiplicative monoid laws and the distributivity and porthole axioms all reduce, after lifting representatives, to ring identities in A closed by ring normalisation; the two involution laws (W7, W8) hold definitionally because pair-swap is its own inverse and commutes with the componentwise product.

Theorem: `WheelFrac.instWheel` (`WheelFrac.lean`)

$\forall \{A : \text{Type}^*\} [\text{CommRing } A] (S : \text{Submonoid } A),$

`Wheel` ($\odot_S A$)

For every commutative ring A and multiplicative submonoid S , the wheel of fractions $\odot_S A$ satisfies all 14 fields of the ZP Wheel typeclass — equivalently, all eight axioms of Carlström's Definition 1.1.

Sorry-free. Lean purity: `[propext, Quot.sound]` — `Classical.choice-free`.

Section IV: Wheel, Not Meadow

A wheel whose two special elements coincide is the one-element wheel (Carlström, Proposition 4.4). For the wheel of fractions the trigger is exactly $0 \in S$: taking $s = 0$ and $s' = 1$ as the witnesses identifies the fraction $[1,0]$ with the fraction $[0,0]$, and every other pair with them. What is proved below is one direction — $0 \notin S$ keeps them apart. The converse is Carlström's own observation that if $0 \in S$ then the relation is improper and $\odot_S A$ is trivial (p. 5). The natural hypothesis $0 \notin S$ holds whenever S is the complement of a prime ideal, the usual case.

Proposition: `WheelFrac.inf_ne_bot` (`WheelFrac.lean`)

$\forall \{A : \text{Type}^*\} [\text{CommRing } A] (S : \text{Submonoid } A),$

$(0 : A) \notin S \rightarrow \infty \neq \perp \text{ in } \odot_S A$

Proposition: WheelFrac.inf_ne_bot (WheelFrac.lean)

When $0 \notin S$, the reciprocal of zero ($\infty = /0$) and the absorbing element ($\perp = 0 \cdot /0$) are distinct. The construction is therefore a non-trivial wheel.

Proof: $\infty = \perp$ would yield witnesses $s, s' \in S$ with $s \cdot 1 = s' \cdot 0$, forcing $s = 0$ and hence $0 \in S$ — contradicting the hypothesis.

Sorry-free. Lean purity: [propext, Quot.sound] — Classical.choice-free.

Remark — What a meadow actually is

A meadow answers the same question a different way, and it adjoins no element at all: Bergstra, Hirshfeld and Tucker define one as "a commutative ring with a total inverse operator satisfying two equations which imply $0^{-1} = 0$." The carrier is unchanged, so there is no pair of new elements available to identify. Bergstra and Ponse fix the default: "by default a 'meadow' is assumed to be an involutive meadow", the involutive ones being exactly those with $0^{-1} = 0$.

A common meadow is the other variant: it adjoins one element — an absorbing element, written a , serving as the inverse of zero. Bergstra and Ponse record what that costs the inverse operation: "the inverse function of a common meadow is not an involution because $(0^{-1})^{-1} = a$," and, setting the two constructions side by side, "wheels are involutive whereas common meadows are non-involutive."

Both directions of why no carrier is at once a non-trivial wheel and an involutive meadow are carried as machine-checked examples in ZeroParadox/Algebra/Wheel.lean §V, with controls showing both hypotheses are load-bearing. The statements are scoped there by their binders, which is why they are cited rather than restated here. Searched 2026-09-13 over the sources listed below: no structural connection between the two constructions located; Bergstra and Ponse report not having found one for common meadows.

Sources: J. A. Bergstra, Y. Hirshfeld and J. V. Tucker, "Meadows and the equational specification of division", arXiv:0901.0823 (abstract); J. A. Bergstra and A. Ponse, "Division by Zero in Common Meadows", arXiv:1406.6878v4, §1 (p. 2) and §4 (p. 14). The concept originates with J. A. Bergstra and J. V. Tucker, "The rational numbers as an abstract data type", JACM 54(2), 2007.

Remark — The Porthole Connection

In ZP-J, the porthole is the point where $\text{val}(\perp) = \infty$ and $\perp = \{\perp\}$ coincide — the same structural fact written in the 2-adic valuation ($v_2(0) = \infty$) and in ZF+AFA (the Quine atom). The wheel of fractions is the algebraic face of that point: $/0$ is defined and distinct from the absorbing $\perp$, exactly the behaviour the porthole predicts. The concrete carrier ZPWheelElem (Wheel.lean §III–VI) makes this explicit on the rationals extended with ∞ and $\perp$, where $\text{val}(x) = \infty \Leftrightarrow /x = \infty$ is proved directly (zpw_top_val_iff_inv_is_inf). This addendum's headline result is the general construction over an arbitrary commutative ring; the concrete carrier is the illustrative special case.

Section V: Scope, Purity, and Relationship to Carlström

Two scope boundaries are worth stating plainly. This is a formalisation, not new mathematics. The wheel of fractions and the theorem that it is a wheel are Carlström's. What is contributed here is a machine-checked encoding faithful to Definition 1.1, with a verified axiom footprint, placed in the ZP porthole context. Ring structure is an input, not a conclusion. The construction starts from a commutative ring and a submonoid; it

does not derive wheel structure from the ZP lattice axioms alone. The bridge typeclass that would state such a derivation, `WheelValuationStructure` (`Wheel.lean` §VII), is defined but its porthole condition $\text{val}(0) = T$ is an assumed axiom, motivated by the ZP argument rather than type-checked as necessary.

Lean Source Files

`WheelFrac.lean` — `rel`, `srel`, the five quotient operations (`waddF`, `wmulF`, `winvF`), `instWheel`, `inf_ne_bot`. The headline results of this addendum.

`Wheel.lean` — the `Wheel` typeclass (`Carlström` Def 1.1, 14 fields), the derived elements `wheelInf` / `wheelBot`, the concrete carrier `ZPWheelElem`, and the porthole correspondence `zpw_top_val_iff_inv_is_inf`.

Both files in `ZeroParadox/` in the public repository.

Axiom Footprint (headline results: `instWheel`, `inf_ne_bot`)

`[propext, Quot.sound]` — `Classical.choice-free`.

`propext` — propositional extensionality (standard in Lean 4)

`Quot.sound` — quotient soundness (standard in Lean 4; the construction is a quotient, so this is expected and unavoidable)

No `Classical.choice`. No Dependent Choice. No set-theoretic assumptions.

This scope is the headline results, not the whole document. Measured against the corpus, the concrete-carrier theorems `zpw_inv_zero_eq_inf`, `zpw_zero_mul_inf_eq_bot` and `zpw_top_val_iff_inv_is_inf` each carry `Classical.choice`; `zpw_inf_ne_bot` is `[propext]`.

Remark R-J.W — Relationship to Carlström's Theorem

The name and the original construction are Setzer's, not Carlström's. Carlström records the lineage on his own p. 3: Edalat and Potts adjoined $\infty = 1/0$ and $\perp = 0/0$ to the reals; Martin-Löf proposed building them into the construction of the rationals from the integers; "such structures were called 'wheels' (the term inspired by the topological picture $\odot$ of the projective line together with an extra point $0/0$) by Setzer [Set97], who showed how to modify the construction of fields of fractions from integral domains so that wheels are obtained instead of fields."

Carlström is the generalizer: "in this paper, we generalize Setzer's construction, so that it applies not only to integral domains, but to any commutative semiring." That is `Wheels` — On Division by Zero (Research Reports in Mathematics No. 11, Department of Mathematics, Stockholm University, 2001; a Licentiate thesis), where Definition 1.1 (p. 5) gives the eight wheel axioms and the wheel of fractions is constructed (§1.2, §4.2). The work was later published as `Wheels` — on division by zero, *Mathematical Structures in Computer Science* 14(1):143–184, 2004. His theorem covers commutative semirings; the Lean encoding in `ZeroParadox/Algebra/WheelFrac.lean` is stated for commutative rings, so it is the narrower case of his result, not the same generality. The result here is that theorem, encoded in Lean 4 against a typeclass that reproduces his Definition 1.1 field for field, and discharged without the axiom of choice. The encoding is what is new: a third party can read the 14 fields against Carlström's eight axioms and confirm the correspondence, and can read the axiom footprint to confirm the choice-free claim. Whether the porthole condition $\text{val}(0) = T$ can be derived — rather than assumed — from upstream ZP structure remains the open question flagged in `Wheel.lean` §VII–VIII.

Endnote: This document is an addendum to ZP-J Self-Reference and reads after it. ZP-J established the porthole ($\text{val}(\perp) = \infty, \perp = \{\perp\}$); this document gives its algebraic form, the wheel of fractions, and the machine-verified proof that the construction is a wheel and a non-trivial one. All results sorry-free in Lean 4 as of September 2026. Footprints are per result, and [propext, Quot.sound] is the headline results'; the Axiom Footprint box above gives the scope and the measured exceptions.